

FPGA-Based Secure Door Lock System

Verilog RTL Design and ModelSim Simulation

Project Type: Digital/RTL Design

HDL: Verilog **Simulation Tool:** ModelSim

Abstract

This project presents a password-based secure door-lock controller designed using Verilog HDL. The system uses a finite state machine (FSM) to control password setting, password checking, successful authentication, failed authentication, and lockout operation. The design incorporates a failed-attempt counter, automatic lockout after three consecutive incorrect passwords, a configurable lockout timer, and a system cycle counter. Four 7-segment display outputs provide visual status indication. The RTL is intended for FPGA implementation and is simulated using ModelSim.

1. Introduction

Digital access-control systems require a controller that can distinguish valid and invalid authentication attempts and respond safely to repeated failures. This project implements such a controller using Verilog RTL. The design stores a 12-bit password, accepts a 12-bit entered password through four 3-bit inputs, compares the entered value with the stored password, and controls the lock output according to the authentication result.

2. Objectives

- Design an FSM-based password authentication controller.
- Provide a password-setting operation.
- Detect incorrect password attempts.
- Limit authentication to three consecutive incorrect attempts before lockout.
- Implement a configurable lockout period using a clock-cycle counter.
- Provide a system cycle counter for timing observation and analysis.
- Display system status using four 7-segment display outputs.
- Verify the RTL behavior through ModelSim simulation.

3. System Architecture

The main RTL components are the FSM controller, password storage/comparator, failed-attempt counter, lockout timer, system cycle counter, and 7-segment display output logic.

Block	Purpose
FSM controller	Controls authentication and lockout sequence
Password register/comparator	Stores and checks the 12-bit password
Attempt counter	Tracks consecutive failed attempts

Lockout timer	Counts the configured lockout duration
System cycle counter	Counts elapsed clock cycles
7-segment logic	Displays SETP, OPEN, FAIL and LOCK status

4. FSM Design

The controller uses six states: IDLE, SETP, CHECK, OPEN, FAIL, and LOCKOUT. IDLE waits for a set or test request. SETP stores the entered password. CHECK compares the entered password with the stored password. OPEN represents successful authentication. FAIL represents an incorrect password, while LOCKOUT disables normal access until the lockout timer expires.

State	Function
IDLE	Wait for set or test command
SETP	Store entered password
CHECK	Compare entered and stored password
OPEN	Successful authentication; unlock asserted
FAIL	Incorrect authentication; increment failure count
LOCKOUT	Prevent authentication until timeout

5. Password Authentication

The four 3-bit inputs p0, p1, p2, and p3 are concatenated into a 12-bit entered password. When the controller is in SETP, the entered value is stored in the password register. During CHECK, the entered password is compared with the stored password. A match causes OPEN, while a mismatch causes FAIL.

```
entered = {p0, p1, p2, p3}
```

6. Failed-Attempt Counter and Lockout

A 2-bit attempt counter tracks incorrect authentication attempts. The third consecutive incorrect password causes the controller to enter LOCKOUT. A successful authentication clears the failed-attempt count. While in LOCKOUT, the unlock output remains inactive and normal password checking is prevented.

7. Lockout Timer

The lockout timer is implemented using lockout_count. LOCKOUT_CYCLES defines the lockout duration in clock cycles. A small value such as 20 is convenient for ModelSim simulation. The counter increments while the FSM is in LOCKOUT and the controller returns to IDLE when the configured duration is reached.

```
parameter integer LOCKOUT_CYCLES = 20;
```

8. System Cycle Counter

A system cycle counter is implemented as a clock-driven register that increments on each active clock edge and is cleared by reset. It provides a measure of elapsed clock cycles from the last reset and is useful for timing analysis and debugging in ModelSim.

```
reg [31:0] cycle_count;
```

9. Seven-Segment Display

The design drives four 7-segment outputs. The current display logic provides status indications for SETP, OPEN, FAIL, and LOCK. The unlock output is asserted only when the FSM is in OPEN.

10. Reset Operation

The reset input is asynchronous and returns the FSM to IDLE. It also clears the failed-attempt counter and lockout timer. The cycle counter is likewise cleared by reset when included in the sequential logic.

11. ModelSim Simulation

The RTL should be simulated in ModelSim using scenarios that demonstrate password setting, correct authentication, incorrect authentication, three consecutive failures, lockout, lockout timeout, and reset. Recommended waveform signals include clk, res, state, attempt_cnt, lockout_count, cycle_count, test, entered, and unlock.

Scenario	Expected result
Correct password	CHECK → OPEN → IDLE; unlock asserted in OPEN
First wrong password	FAIL; failure count increases
Second wrong password	FAIL; failure count increases
Third wrong password	LOCKOUT begins
During lockout	Unlock remains inactive
Lockout timeout	LOCKOUT → IDLE
Reset	FSM and counters return to initial values

12. Tools and Technologies

- Verilog HDL — RTL design and implementation
- ModelSim — compilation, simulation, and waveform analysis
- FPGA-oriented digital design concepts — FSMs, counters, registers, and display control

13. Conclusion

The project demonstrates the RTL design of a password-based digital access-control system using Verilog. The FSM provides the core control sequence, while the failed-attempt counter and timed lockout improve protection against repeated incorrect attempts. The seven-segment display provides visible system status, and the system cycle counter provides timing information for simulation and analysis. The project demonstrates practical application of FSMs, counters, registers, password comparison, and combinational output logic in an FPGA-oriented RTL design.

14. Future Enhancements

- Add memory-based event logging for security events.
- Add a circular buffer to retain the most recent events.
- Develop a separate SystemVerilog verification project with assertions and functional coverage.

Project scope note: This report intentionally excludes UART, multi-user authentication, programmable passwords, and a SystemVerilog testbench from the door-lock project.